

Technical Modeling Report — Financial Distress Prediction (Stage121)

Models compared on an identical feature set: **Logistic Regression, Random Forest, XGBoost**. Main metric: **PR-AUC**. Forecast horizon: **one year ahead** (firm info at year $t \rightarrow$ distress at $t+1$).

1. Data & target

- One-year-ahead rows built: **983**, positives **37** (3.76%).
- Only dropped predictors: {'predictor_last_year_no_next_panel_row': 128, 'next_year_gap_no_row': 0, 'next_target_invalid': 0}.
- Target distressed_target_reviewed was never modified; year- t label is excluded from features.

2. Temporal split (approved revision)

Distress events exist only in target years 1393–1398, so the brief's original test window (1401–1402) contained a single positive and two validation folds with zero positives. With written approval the boundary was moved earlier: - **dev**: n=300, positives=27 - **test**: n=200, positives=9 - Rows with target_year $\geq$ 1399 (zero positives, after the test window) are excluded from one-year-ahead modeling and documented.

3. Pipeline

- All preprocessing (median impute + missing indicators, RobustScaler for LR, one-hot industry) is fit on the training fold only — no leakage.
- Imbalance handled by cost-sensitive learning (class_weight / scale_pos_weight); SMOTE only in robustness.
- Hyperparameters tuned with Optuna on expanding-window folds; selection metric = mean validation PR-AUC. Thresholds chosen on validation only.

4. Cross-validation (validation folds)

model	po	pr	roc	brier
logistic	17811	0.023	0.717	
logistic	23433	0.015	0.759	
logistic	37533	0.024		
random_forest	17811	0.037	0.717	
random_forest	23433	0.033		
random_forest	37009	0.031		
xgboost	17827	0.026		
xgboost	24033	0.023		
xgboost	37009	0.023		
## 5. Test results (optimal threshold, primary = F1)				
model	po	pr	roc	f1f2
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logistic	0.263	0.025	0.732	
random_forest	0.633	0.025	0.726	
xgboost	0.623	0.027	0.733	
## 6. Test 95% cluster-bootstrap CIs (by ticker)				
metric	n	po	pr	roc
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pr_auc	0.263	0.025		
roc_auc	0.753	0.025		
brier	0.123	0.025		
precision	0.267	0.025		
recall	0.443	0.025		
f1	0.325	0.025		
f2	0.391	0.025		
specificity	0.901	0.025		

[illegible]

B_extended	logb0.3371.2
A_winsorized	logb0.2371.6
A_smote	logb0.211.32
B_drop_near_definition	logb0.25237
same_year_classification	logb0.2556.5
A_main	xgb0.51157.1
B_extended	xgb0.51235.4
A_winsorized	xgb0.35357.1
A_smote	xgb0.311435
B_drop_near_definition	xgb0.35567.6
same_year_classification	xgb0.353452
## 10. Documented data limitations (brief)	
- 39 missing operating_cash_flow (1 source_unavailable_codal, 38 deferred); 145 unresolved gross_profit; 13 incomplete provenance; 99 unreviewed abnormal financial changes. None were zero-filled.	
## 11. Caveats	
- Only 9 positives in the test window → test metrics and bootstrap CIs are wide; PR-AUC on validation folds is the more stable selection signal.	